

FULL-STACK AI / FORWARD-DEPLOYED ENGINEERING INTERNSHIP

Weekend Technical Assessment

Atlas Fresh — Daily Apple Export Planner

Format	Qarizmi candidate brief
Timebox	10–12 hours maximum; stop and state what remains
Submission	Repository, concise README and 3–5 minute walkthrough
Technology	Your choice; use the stack you can ship and explain
Data	One supplied workbook; all farms, clients and figures are fictional

The goal: Build the clearest possible workspace for the daily Production–Commercial committee: compare what farms were expected to produce with what arrived, decide which apples to export to which clients, and make the low-value local-market residual impossible to miss. User experience is the first thing we assess.

1. Business context

Atlas Fresh in one minute

Atlas Fresh is a fictional apple producer and exporter. It coordinates 20 farms, one export-conditioning station and 10 export clients. The apples are classified into four quality segments: A is the highest quality, followed by B, C and D. Each client buys a maximum number of tonnes, accepts a defined quality rule and pays a price per tonne.

Business element	Case	Business element	Case
Product	Apples	Production network	20 farms
Quality	Segments A / B / C / D	Daily station capacity	500 t
Commercial program	10 export clients	Fallback	Local market at 10% of segment reference price

How the season is planned

Before the season starts, Commercial builds a sales program for the 10 clients: daily tonnes, accepted quality and export price. In parallel, Production sets an expected daily capacity and an expected mix of Segments A, B, C and D for each farm. These are two separate plans: no farm is preassigned to a client in this case.

Step	Season process	Output
1	Commercial commits	Client demand, segment rule and export price.
2	Production plans	Expected daily capacity and A/B/C/D mix for each farm.
3	Farms receive targets	What the business expects each farm to deliver by quality segment.
4	The station prepares	People and equipment are planned around the commercial and farm program.

This program is essential, but it is only a forecast. Weather, maturity, size and visual quality change the quantity and the segment mix that each farm actually delivers.

What happens every day

On the day represented by the supplied workbook, the farms were expected to deliver 600 t but actually delivered 560 t. The gap is not only volume: Segment A—the hardest quality to produce—is 11.7 t below plan. Segments B and C are also below plan, while Segment D is above plan. Meanwhile, the export-conditioning station can process only 500 t for export.

Any apple not exported is sold through the local channel. In this simplified case, that fruit is worth only 10% of the reference export price for its quality segment. A tonne left for the local market therefore destroys most of its potential commercial value.

How the decision is made today

Production and Commercial hold a daily planning committee. Production brings the latest farm receipts, segment mix and station capacity. Commercial brings client demand, accepted qualities and prices. The teams currently compare spreadsheet tables line by line, discuss reallocations and rebuild totals manually whenever an input changes.

Perspective	Daily question
Production	Which farms and segments are above or below plan? What can the station execute?
Commercial	Which clients can be complete, partial or unserved? Which exports protect the most value?
Joint decision	Which farm-segment volume serves which client, and what falls back to the local market?

Current pain: time is spent reconstructing the situation, while calculations and explanations are difficult to reproduce consistently.

The decision Atlas Fresh wants to automate

The operating goal is to export as much of the actual crop as possible. For this exercise, do not build a generic mathematical optimizer: apply the authoritative deterministic policy below. It protects higher-priced orders first, then allocates as much compatible volume as possible until client demand or station capacity is reached. Export rate is a key business outcome, not a separate optimization algorithm.

- Which farms missed or exceeded their expected capacity and segment mix?
- Which client orders are complete, partial or unserved, and why?
- Exactly which farm and segment serves each client?
- How much of the actual crop is exported, and how much goes local at a steep discount?
- What station capacity, export revenue, local value and total value result from the plan?

Human-in-the-loop boundary: This is a decision-support product. It prepares the committee; it does not contact farms or clients, confirm the operating plan, or write to an external system. Production and Commercial remain responsible for approving execution.

The product succeeds when the committee spends less time asking ‘What happened?’ and more time deciding which exceptions and trade-offs to approve.

2. What you are building

Build one browser-based Production–Commercial planning workspace for the supplied daily snapshot. It may use tabs or sections, but it should feel like one coherent product rather than a collection of technical screens.

Without your narration, a non-technical manager should understand the situation in under one minute and trace any allocation or shortage in under three minutes.

The five-step user journey

Step	Action	Outcome
1	Load	Load and validate the supplied workbook on the server.
2	Compare	See expected versus actual tonnes by farm and quality segment.
3	Plan	Generate the deterministic export allocation on the server.
4	Decide	Understand client service, station use, export value and local residual.
5	Explain	Ask a grounded assistant to explain the calculated plan with evidence IDs.

What success feels like

- Production sees plan-versus-actual gaps by farm and segment, plus station utilization.
- Commercial sees every client as complete, partial or unserved with a useful reason.
- Any allocation can be traced to a farm ID, segment, client ID, tonnes and revenue.
- The 60 t local residual is visible by farm and segment, with its value impact.
- The interface handles invalid data and server failure without losing user trust.

3. Data and deterministic planning policy

Use `Atlas\_Fresh\_Production\_Commercial\_Data.xlsx`. It contains four sheets:

Sheet	Meaning
Read Me	Business rules and public baseline checks
Farms	20 farm production plans: expected capacity/mix and actual A/B/C/D receipts; no client assignment
Clients	10 export orders: quality rule, demand and export price
Station	500 t capacity, 10% local-market ratio and reference export price by segment

Daily snapshot only: The workbook deliberately contains no farm-to-client mapping. Producing that traceable daily mapping is part of the assignment. A farm may serve several clients, and a client may receive apples from several farms or compatible segments. You are not being asked to forecast or optimize the full season.

Quality compatibility

Rule	Meaning	Example
EXACT	Only the requested segment	C03 EXACT B accepts B only
MINIMUM	The requested segment or a better segment	C04 MINIMUM B accepts B or A
Quality order	A is highest, followed by B, C and D	For MINIMUM C, prefer C before B or A

Reference policy — implement this exact order

1. Validate IDs, acceptance modes, segments and the complete segment reference-price table. Expected mix cells are decimal fractions from 0 to 1 and must sum to 1.0 per farm.
2. Validate quantities precisely: actual A/B/C/D tonnes, client demand and station capacity are non-negative multiples of 5 t; expected daily capacity may use one decimal.
3. Create the available supply from each farm's actual A/B/C/D tonnes. Planned values are for comparison, not allocation.
4. Process client orders by export price per tonne descending; break a price tie by client_id.
5. For the current client, keep only compatible farm-segment supply with a positive balance.
6. Sort compatible supply by the smallest quality upgrade, then farm_id. This is the deterministic rule that creates the farm-to-client assignment.
7. Allocate in 5 t steps until client demand, compatible supply or station capacity is exhausted.
8. After all clients, send every unexported actual tonne to the local market. Local value equals residual tonnes $\times$ local-market ratio $\times$ the reference export price of the residual fruit's segment. Segment reference prices never affect client order or export revenue.
9. Return allocation rows, farm/segment balances, client statuses, shortage reasons and KPIs. The same input must always return the same result.

Important engineering boundary: Implement the planning policy in deterministic server-side code. Do not ask an LLM to choose farms or clients, calculate quantities, enforce constraints or produce KPIs.

Public baseline checks

Metric	Value	Metric	Value	Metric	Value
Expected plan	600.0 t	Actual received	560.0 t	Capacity	500.0 t
Actual A / B	90.0 / 160.0 t	Actual C / D	180.0 / 130.0 t	Export	500.0 t
Export rate	89.3%	Local volume	60.0 t	Export revenue	EUR 549,500
Local value	EUR 4,500	Total value	EUR 554,000	At-risk clients	3

KPI definitions: Expected segment tonnes = expected daily capacity × expected segment percentage. Segment variance = actual – expected. Export rate = exported tonnes ÷ total actual tonnes. Export revenue = Σ allocated tonnes × served client price. Total value = export revenue + local value. A client is COMPLETE when allocation equals demand, PARTIAL when allocation is between zero and demand, and UNSERVED when allocation is zero. At risk means partial or unserved. Evaluators will change a valid input, so every check must be calculated.

4. Mandatory product requirements

4.1 Server-side input and validation

- Load the supplied workbook through a server route. A preloaded seed action is enough; a complex upload interface is not required.
- Reject duplicate or missing IDs, missing segment reference prices, invalid modes or segments, mix fractions outside 0–1 or not totalling 1.0, negative quantities, non-5 t actual/demand/capacity values and invalid capacity.
- Show an actionable message that identifies the sheet and farm/client ID involved. Never repair invalid input silently.
- Keep the source workbook unchanged and separate source data from computed results.

4.2 Deterministic daily planning engine

- Run the reference policy on the server and return farm-client-segment allocation rows, client status, residual supply and KPIs.
- Give every partial or unserved client a reason. If capacity is exhausted, use `STATION\_CAPACITY\_REACHED`; otherwise use `INSUFFICIENT\_COMPATIBLE\_SEGMENT` when compatible supply is insufficient.
- Calculate and test these invariants: export $\leq$ station capacity; client export $\leq$ demand; farm-segment export $\leq$ actual; every allocation is compatible; export + local = actual received.
- The baseline must match the public checks and all outputs must change when a valid input changes. Hard-coded baseline results are not accepted.
- Include four to six meaningful automated tests covering ordering, compatibility, hard limits, local residual and validation.

4.3 One excellent planning workspace

- Lead with the business decision: data health, planned versus actual, export capacity, export rate, local volume, export revenue and total value.
- Production view: 20 farms, expected capacity/mix, actual A/B/C/D, useful variances and residual local tonnes.
- Commercial view: client rule, demand, allocated, remaining, revenue, status and shortage reason.
- Allocation detail: farm ID, segment, client ID, tonnes, quality upgrade if any, and export revenue.
- Make the connection between a farm variance and a client risk understandable; do not present disconnected tables only.
- Provide useful loading, empty, validation-error and server-error states, plus a predictable retry or reset path.
- Design for keyboard use and real work at 1024 px and 1440 px. Charts, drag-and-drop and multiple pages are optional.

4.4 Grounded planning assistant

Add one small assistant panel that explains the structured result produced by your engine. Implement at least one real hosted or local model path when configured. Candidates are not expected to purchase API access: without a key or local model, show an honest no-key state and a clearly labelled deterministic summary.

- Support three questions: Which clients are at risk and why? Which farm/segment gaps matter most today? Why are 60 t going local and what is their estimated value?

- Every number must come from the server result and every answer must cite resolvable client IDs, farm IDs or segment labels.
- Send only the minimum structured context required. Validate model output and remove or reject unknown IDs.
- If the answer is not present in the inputs or computed plan, say that it is unavailable.
- Handle no-key, timeout and invalid-output states honestly; never show a fake AI answer.
- Include one grounded-answer check and one unsupported or provider-failure check.

Read-only AI: The assistant may explain the plan. It must not calculate or change allocations, confirm execution, or write external data.

5. Acceptance checklist

ID	Area	Pass condition
A	Inputs	The app shows 600 t expected, 560 t actual, A/B/C/D totals and 500 t capacity.
B	Baseline	It calculates 500 t export, 60 t local, 89.3% export rate and the exact public values.
C	Clients	C02 and C09 are partial for segment shortage; C08 is partial because station capacity is reached.
D	Trace	Every exported tonne resolves to one valid farm, segment and client; every residual resolves to local.
E	Limits	No demand, farm-segment balance or station capacity is exceeded.
F	Validation	Invalid IDs, modes, segments, mix, quantities or capacity are rejected server-side.
G	Assistant	Supported answers cite real IDs; unsupported questions and provider failure are honest.
H	UX	A manager can identify gaps, at-risk clients and local impact without candidate narration.
I	Reproducible	A clean clone starts, tests and builds using only documented commands.

6. Delivery

- A Git repository with source code and meaningful commit history.
- A concise `README.md`: prerequisites, one clean-start path, test/build commands, architecture, assumptions, limitations and the next three production steps.
- Automated tests for the planning policy, constraints, validation and assistant boundaries.
- A 3–5 minute walkthrough aimed at Production and Commercial users.

- A short note listing AI coding tools used, what you verified, approximate time spent and intentional omissions.
- Optional live URL. It never replaces clean-clone reproducibility.

Explicit non-requirements

- No authentication, roles, audit workflow or database persistence.
- No full-season forecasting, ML training, vector database, RAG corpus or autonomous action.
- No logistics, freight, weather, packaging, variety, calibre, ERP, email or messaging integration.
- No multi-product, multi-station, multi-day optimizer, manual allocation editor or capacity scenario UI.
- No microservices, Kubernetes, event bus, mobile app or spreadsheet clone.
- No paid service. Docker, CI and a live deployment are bonuses, not requirements.
- No hidden cleaning puzzle. The supplied workbook is a clean, authoritative baseline.
- This is a synthetic recruitment exercise. Do not use real client data or build a client production deliverable.

Evaluation rubric

Area	Points	What we assess
User experience and product judgment	40	Clarity, task speed, Production-Commercial connection, feedback, accessibility and useful trade-offs.
Data and planning correctness	25	Exact policy, compatibility, hard limits, reasons, baseline, local value and no hard-coding.
Full-stack engineering and tests	15	Server validation, API design, architecture, errors, types, tests and build.
Grounded AI assistant	10	Correct boundary, evidence IDs, minimal context, validation and honest fallback.
Delivery and ownership	10	Clean start, README, walkthrough, AI disclosure and honest scope.
Total	100	The mandatory user journey is worth more than optional infrastructure or visual effects.